

Model for seasonal variation

Parametrization

A model for seasonal variation with periodicity m for the random vector $\mathbf{x} = (x_1, \dots, x_n)$, $n > m$ is obtained assuming that the sums $x_i + x_{i+1} + \dots + x_{i+m-1}$ are independent Gaussian with precision τ .

The density for $\mathbf{x}$ is derived from the $n - m + 1$ increments as

$$\pi(\mathbf{x}|\tau) \propto \tau^{\frac{(n-m+1)}{2}} \exp \left\{ -\frac{\tau}{2} \sum (x_i + x_{i+1} + \dots + x_{i+m-1})^2 \right\} \quad (1)$$

$$= \tau^{\frac{(n-m+1)}{2}} \exp \left\{ -\frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} \right\} \quad (2)$$

where $\mathbf{Q} = \tau \mathbf{R}$ and $\mathbf{R}$ is the structure matrix reflecting the neighbourhood structure of the model.

Hyperparameters

The precision parameter τ is represented as

$$\theta = \log \tau$$

and the prior is defined on θ .

Specification

The seasonal model is specified inside the `f()` function as

```
f(<whatever>, model="seasonal", season.length=<season.length>,  
    scale.model=<scale.model>, hyper = <hyper>)
```

Hyperparameter specification and default values

`doc` Seasonal model for time series

`hyper`

`theta`

```
hyperid 7001  
name log precision  
short.name prec  
prior loggamma  
param 1 5e-05  
initial 4  
fixed FALSE  
to.theta function(x) log(x)  
from.theta function(x) exp(x)
```

`constr` FALSE

`nrow.ncol` FALSE

`augmented` FALSE

`aug.factor` 1

```
aug.constr
n.div.by
n.required FALSE
set.default.values FALSE
missing.values.warning TRUE
pdf seasonal
```

Example

```
n=203
n.seas=12
trend=1:n
seasonal.sim=rep(1:n.seas, ceiling(n/n.seas))[1:n]

a=1
b=0.5
y = rnorm(n,a+b*trend,1)+rnorm(n,0.2*seasonal.sim,1)

data=data.frame(y=y,trend=trend,seasonal=trend)
formula = y~f(trend,model="rw2")+f(seasonal,model="seasonal",
                                   season.length=n.seas)
result=inla(formula,family="gaussian",data=data)
```

Notes

The seasonal is intrinsic with rank deficiency $m - 1$.